FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA

National Fuel Pass System (NFPS)

ETHIO FUEL PASS

Software Requirements Specification

Fair. Controlled. Transparent.
Fuel for a Stronger Ethiopia.

Prepared by

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Addis Ababa, Ethiopia

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1. Document Control

This document specifies the functional, non-functional, architectural, and user-experience requirements for the Ethio Fuel Pass System (NFPS). It is the binding scope reference for the MVP build (Phase 1 Pilot, 0–6 months) and the production-grade scale-out that follows (Phase 2–3, 6+ months).

Field	Value
Document Title	Ethio Fuel Pass — Software Requirements Specification
Document Owner	Eldix IT Technology PLC (360Ground™)
Author	Biruk Hailu, Founder & CEO
Version	1.0 (Draft for Review)
Audience	Government sponsor, product, engineering, QA, security, operations
Backend Stack	Python 3.12 / Django 5.x / Django REST Framework / PostgreSQL 16
Client Channels	Mobile (Android/iOS), Web Portal, USSD/SMS, Service Centers, Station Operator App
Confidentiality	Restricted — Government / Project Stakeholders only

1.1 Revision History

Version	Date	Author	Summary
0.1	TBD	B. Hailu	Initial outline from concept deck.
0.5	TBD	B. Hailu	Functional modules, user stories, acceptance criteria.
1.0	TBD	B. Hailu	Architecture (MVP + Production), NFRs, sign-off draft.

2. Executive Summary

Ethiopia's fuel distribution faces persistent, well-documented problems: frequent shortages, inequitable access, hoarding and black-market diversion, leakage and wastage, and a structural absence of real-time data for planners. The Ethio Fuel Pass System (NFPS) addresses these by giving every eligible vehicle owner a weekly fuel quota, delivered as a unique, time-bound QR Code (the "Fuel Pass") and validated at the pump in real time.

The platform binds three identities together — the citizen (via the Fayda National Digital ID), the vehicle (via the Vehicle Registry), and the transaction (via a station POS) — so that one citizen, one vehicle, and one Fuel Pass map cleanly to one fair quota.

2.1 Vision

A digital, transparent, accountable system that ensures every liter of fuel reaches the right user, at the right time, at the right price.

2.2 Strategic Alignment

- Digital Ethiopia 2030 — advances digital public service delivery and inclusive citizen-centric services.
- Smart Governance — strengthens transparency, anti-corruption, and evidence-based policy.
- Digital ID Utilization — extends the value of Fayda by binding it to a high-value daily-life service.
- Data-Driven Policymaking — produces real-time consumption and demand signals at national, regional, and station level.

2.3 Expected Impact (per concept deck)

KPI	Target Range
Reduction in fuel wastage and theft	20–40%
Improvement in fuel availability	30–50%
Increase in data accuracy & integrity	Up to 300%
Operational cost reduction (distribution side)	25–45%
Enhanced monitoring & planning capability	40–60%

2.4 Delivery Approach

The system is delivered in two engineering tracks, both governed by this SRS:

- MVP — a deliberately minimal Django application with the smallest viable layer count, optimized for rapid pilot in Addis Ababa with private cars and taxis (0–6 months).

- Production — a horizontally-scaled, cache-fronted, queue-backed, observability-rich version of the same Django codebase, hardened for national rollout (6–18 months) and continuous optimization (18+ months).

3. Glossary & Acronyms

Term	Meaning
NFPS	National Fuel Pass System — the platform described by this SRS.
Fuel Pass	A unique, time-bound, signed QR Code representing one weekly fuel quota for one vehicle.
Quota	Volume of fuel (liters) allocated to a vehicle for a given quota period (default: one week).
Quota Period	Default Monday 00:00 to Sunday 23:59 East Africa Time (EAT).
Vehicle Category	Classification used by the quota engine (private car, taxi, public bus, ambulance, government, commercial truck).
Fayda	Ethiopia's National Digital ID; canonical citizen identity used for KYC.
Station	A licensed fuel retail outlet enrolled on NFPS.
Operator App	The Android application used by a station attendant to scan and authorize a Fuel Pass.
RBAC	Role-Based Access Control — back-office permission scheme.
KYC	Know Your Customer — citizen and vehicle verification at registration.
EAT	East Africa Time (UTC+3).
ETB	Ethiopian Birr.
MVP	Minimum Viable Product.
NFR	Non-Functional Requirement.
SLA	Service Level Agreement.

4. Stakeholders & Personas

Five primary personas drive the System. Each has a dedicated channel set, a constrained mental model, and a measurable success criterion. The product must be designed for the lowest-bandwidth member of each persona group.

4.1 Persona 1 — Citizen Driver (Almaz)

Attribute	Detail
Profile	Private vehicle owner in Addis Ababa, mid-30s, owns a 2015 sedan.
Devices	Mid-range Android phone, 4G/3G, intermittent connectivity.
Languages	Amharic (primary), some English.
Pain today	Long fuel queues, uncertainty about availability, suspicion that quota is unfairly distributed.
Goal	Know weekly quota in advance, get fuel without queueing, trust the process.
Success metric	Reaches a station and dispenses fuel in ≤ 3 minutes from arrival.

4.2 Persona 2 — Commercial Driver (Tesfaye, Taxi)

Attribute	Detail
Profile	Code 02 taxi driver, 40s, drives 12+ hours/day; fuel is his largest cost.
Devices	Low-end Android, often on USSD.
Languages	Amharic only.
Pain today	Quota too low for working hours, time wasted in queues.
Goal	Category-appropriate quota, quick redemption, predictable refill rhythm.
Success metric	Quota fits real route demand; redeems via USSD if app fails.

4.3 Persona 3 — Station Operator (Hana, Pump Attendant)

Attribute	Detail
Profile	20s, trained for one shift, handles 200+ vehicles/day.
Devices	Station-issued Android tablet/phone (Operator App).

Attribute	Detail
Pain today	Cash handling, manual logbooks, frequent disputes.
Goal	Scan QR, get GREEN/RED in under 1.5 seconds, dispense, record, move on.
Success metric	Average transaction time ≤ 30 seconds; near-zero dispute rate.

4.4 Persona 4 — Government Administrator (Mekonnen, Ministry of Energy)

Attribute	Detail
Profile	Senior policy officer; sets quota policy and reviews national consumption.
Devices	Desktop, modern Chrome / Edge.
Pain today	No real-time picture, monthly aggregated reports, stale data.
Goal	Adjust quota policy by category and region, see live consumption dashboards, run forecasts.
Success metric	Time-to-policy-change measured in hours, not weeks.

4.5 Persona 5 — Auditor / Inspector (Eden, Internal Audit)

Attribute	Detail
Profile	Reads transaction histories, investigates anomalies.
Devices	Desktop web portal.
Pain today	Paper trails, no end-to-end visibility from depot to dispenser.
Goal	Trace any liter of fuel from allocation to dispensing with one click.
Success metric	End-to-end trace of any single transaction in ≤ 60 seconds.

5. System Scope

Scope is split into two distinct releases. The MVP exists to validate the core fuel-pass loop in pilot conditions; the Production release exists to operate the system at national scale, with full resilience and integrations.

5.1 In-scope (MVP, Phase 1, Months 0–6)

- Citizen self-registration via the mobile app and web portal, with manual KYC fallback at service centers.
- Vehicle registration tied to a citizen, with category assignment by an admin.
- Quota policy with one default rule per vehicle category, set by a single national admin.
- Weekly quota allocation engine, run by a scheduled job at 00:00 EAT every Monday.
- Fuel Pass (signed QR) generation, available in-app and as an SMS link.
- Station Operator App for Android: QR scan, online validation, dispense confirmation, transaction log.
- Online-only validation for the MVP (no offline mode in the station app).
- Basic admin web portal: users, vehicles, stations, categories, quota policy, transaction search.
- Read-only national dashboard: liters allocated, liters dispensed, top stations, top categories.
- Audit log of every state-changing action.

5.2 In-scope (Production, Phase 2–3, Months 6–18+)

- Full integration with Fayda for live KYC, with real-time photo and biometric checks.
- Full integration with the National Vehicle Registry for live vehicle data and license status.
- USSD short-code and SMS gateway for low-bandwidth quota check and Fuel Pass retrieval.
- Service Center desktop app for assisted registration.
- Offline-capable Operator App with signed-QR offline validation and a forward-secure replay window.
- Integrated Payments — direct debit / wallet / card, with reconciliation against station settlements.
- Multi-region quota policy with sub-regional overrides (state, zone, woreda).
- Anomaly detection (statistical and ML) on transactions, with case management for auditors.
- AI-driven demand forecasting, refreshed daily.
- Public open-data API (read-only, anonymized) for civic transparency.

5.3 Out-of-scope (this document)

- Hardware procurement specifications for station tablets, scanners, or pump-side IoT meters.

- Subsidy economics, per-liter pricing, and tax modeling.
- Fleet-card replacement programs for state enterprises (covered separately).
- Custodial wallet / mobile-money issuance — NFPS integrates with existing rails, it does not become a payment institution.

5.4 MVP vs Production — at a glance

Capability	MVP	Production
Registration channels	App + Web	App + Web + USSD + SMS + Service Center
KYC	Fayda lookup, manual override allowed	Fayda live verification, biometric, auto-block
Vehicle data	Manual entry + admin verify	Live Vehicle Registry sync
Operator App	Online-only	Offline-capable, signed-QR replay window
Quota policy	Single national rule per category	Region/zone/woreda overrides
Payments	Out-of-band	Integrated rails + reconciliation
Analytics	Read-only static dashboard	Live dashboards + anomaly detection + forecasting
Caching	In-process / per-process LRU	Redis cluster, per-resource TTL
Async work	Django mgmt commands, scheduled (cron)	Celery + Redis broker, retries, dead-letter queue
Files / receipts	Local disk	Object storage (S3-compatible) + CDN
Search	Postgres LIKE / trigram	Postgres + OpenSearch for transactions
Observability	Sentry + Django logging	Sentry + OpenTelemetry + Prometheus + Grafana
Deployment	Single VM (Docker Compose)	Kubernetes, multi-AZ, HPA, blue/green

6. High-Level Architecture

The System is organized into five functional modules sitting on a shared Django code base. Each module corresponds to a Django app, exposes a REST API, emits domain events, and writes to dedicated tables. The same module set ships in MVP and Production; what changes between releases is depth (capabilities), surface (channels), and scale (infrastructure).

6.1 Module Map

#	Module (Django app)	Responsibility
M1	users_vehicles	Citizen accounts, KYC, vehicle records, category & eligibility.
M2	quota	Quota policy, weekly allocation engine, manual adjustments.
M3	fuelpass	Fuel Pass generation, signing, delivery, revocation.
M4	transactions	QR validation, authorization, dispense logging, reconciliation.
M5	analytics	Dashboards, reports, alerts, forecasting (read-side projections).
M0	core	Cross-cutting: auth, RBAC, audit log, settings, health, integrations.

6.2 Channels and Surfaces

- Citizen Mobile App (Android, iOS) — primary channel, offline-tolerant for QR display.
- Citizen Web Portal — registration, profile, history, downloadable receipts.
- USSD / SMS — low-bandwidth fallback (Production).
- Service Center — assisted on-boarding (Production).
- Station Operator App (Android) — scanning, authorization, dispense logging.
- Government Admin Portal — policy, dashboards, audit, RBAC.

6.3 Cross-Module Data Flow (one redemption)

1. M3 issues a Fuel Pass for vehicle V in week W, signed by the platform private key.
2. Citizen presents the QR at a station; M4 receives the scan from the Operator App.
3. M4 verifies signature, expiry, and one-time-use status against the redemption ledger.
4. M4 calls M2 to check remaining quota for V in W.
5. If allowed, M4 reserves the quota, returns AUTHORIZED to the Operator App, and waits for dispense confirmation.
6. On confirmation, M4 commits the dispense, writes the transaction, decrements the quota, and emits a `transaction.completed` event.
7. M5 consumes the event asynchronously and updates the analytics projection.

7. Functional Requirements

Each functional requirement is expressed as a User Story with Acceptance Criteria in Given / When / Then form. IDs follow the convention M<module>-US-<n>.

7.1 Module M1 — Users & Vehicles

M1 owns identity and eligibility. A vehicle that is not registered, not verified, or whose owner failed KYC, cannot receive a Fuel Pass.

M1-US-01 User Story

As **a citizen with a valid Fayda ID**, *I want* register an account on NFPS using my Fayda number and a phone OTP, *so that* I can be uniquely identified by the platform without re-submitting paper documents.

Acceptance Criteria (Given / When / Then)

AC1. Given I open the app and select 'Sign up', when I enter a 16-digit Fayda number and a +251 phone, then the System sends a 6-digit OTP within 30 seconds.
AC2. Given I enter the correct OTP within 5 minutes, when the System validates the Fayda number against the integration (or local mock in MVP), then my account is created and I land on the home screen.
AC3. Given I enter a Fayda already linked to another active account, when I attempt sign-up, then the System rejects the request with error code USR-409 'Identity already registered'.
AC4. Given I exceed 5 failed OTP attempts in 1 hour, when I retry, then the System enforces a 24-hour lockout and writes an entry to the audit log.

M1-US-02 User Story

As **a registered citizen**, *I want* add a vehicle to my account using its plate number and engine/chassis number, *so that* the System can issue Fuel Passes against my vehicle.

Acceptance Criteria (Given / When / Then)

AC5. Given I am authenticated, when I submit plate, region code, engine and chassis numbers, then the System creates a Vehicle in PENDING_VERIFICATION state.
AC6. Given Vehicle Registry integration is active (Production), when I submit, then the System auto-verifies and moves the vehicle to ACTIVE within 60 seconds.
AC7. Given Vehicle Registry is unavailable (MVP), when I submit, then an admin must manually verify within 72 hours; the citizen is notified by SMS on state change.
AC8. Given the plate is already linked to another active citizen, when I submit, then the System rejects with code VEH-409 and shows 'This vehicle is already registered'.

M1-US-03 User Story

As **a government administrator***, I want* assign or change the category of a vehicle (e.g. private, taxi, public bus, ambulance)*, so that* the quota engine can compute the right allocation.

Acceptance Criteria (Given / When / Then)

AC9. Given I have the role `vehicle_admin`, when I open a vehicle and change its category, then the change is saved and an audit entry is written with my user ID and timestamp.

AC10. Given a citizen does not have `vehicle_admin`, when they call the category endpoint, then the System returns 403 with code AUTH-403.

AC11. Given I change a category mid-week, when the next allocation runs, then the new category is used; the current week's already-issued pass is unchanged.

M1-US-04 User Story

As **a citizen***, I want* see and update my profile, phone number, and vehicles*, so that* I can keep my information current without visiting an office.

Acceptance Criteria (Given / When / Then)

AC12. Given I am authenticated, when I view my profile, then I see my name (from Fayda), Fayda ID (masked except last 4), phone, and a list of my vehicles with state.

AC13. Given I change my phone, when I submit, then the System sends OTP to the new number and only commits the change after successful OTP within 5 minutes.

AC14. Given I delete a vehicle, when I confirm, then the vehicle is soft-deleted (state = ARCHIVED), no future passes are issued, and historical transactions remain visible to me and to auditors.

7.2 Module M2 — Quota Management

M2 owns the policy that turns ‘who and what kind of vehicle’ into ‘how much fuel this week’. It is the only place quota numbers exist, and the only place they may be changed.

M2-US-01 User Story
As **a national policy administrator**, *I want* define a quota rule per vehicle category (volume in liters per quota period), *so that* the System has an authoritative source of truth for weekly allocations.
Acceptance Criteria (Given / When / Then)
AC15. Given I am role \`quota_admin\`, when I create a rule with category=PRIVATE_CAR, liters=20, period=WEEKLY, then the rule is saved with a \`valid_from\` of next quota period. AC16. Given a rule exists, when I edit it, then the existing rule is end-dated and a new versioned rule is created (no in-place mutation). AC17. Given two rules overlap on category and time, when I save, then the System rejects with code QUO-409 ‘Overlapping policy version’.

M2-US-02 User Story
As **the System (scheduled job)**, *I want* run the weekly allocation engine at 00:00 EAT every Monday, *so that* every active vehicle has a Fuel Pass before the working week begins.
Acceptance Criteria (Given / When / Then)
AC18. Given the scheduler triggers at 00:00 EAT Monday, when the job runs, then it loads all ACTIVE vehicles, computes their quota from the active rule, and writes one Allocation row per vehicle for the period. AC19. Given a vehicle was archived since last week, when the job runs, then no Allocation is created for it. AC20. Given the job is interrupted mid-run, when it restarts, then it is idempotent: it produces exactly one Allocation per vehicle per period (enforced by a unique constraint on vehicle_id + period_start). AC21. Given the job completes, when it finishes, then it emits a \`quota.allocated\` event consumed by M3 to issue Fuel Passes.

M2-US-03 User Story
As **a quota administrator**, *I want* manually adjust a single vehicle’s allocation for a given week (e.g. +10 L for an ambulance), *so that* exceptional cases can be handled without overhauling policy.
Acceptance Criteria (Given / When / Then)

AC22. Given I have role `quota_admin`, when I add an Adjustment(vehicle, period, delta_liters=+10, reason), then the available quota for that pass becomes original + delta.

AC23. Given the delta would result in a negative quota, when I submit, then the System rejects with code QUO-422 ‘Adjustment would result in negative quota’.

AC24. Given I make an adjustment, when it commits, then the System writes an audit entry containing user, vehicle, period, delta, and free-text reason; reasons must be ≥ 10 characters.

M2-US-04 User Story

As **a citizen**, *I want* see my remaining quota in real time, *so that* I know how many liters I can still buy this week.

Acceptance Criteria (Given / When / Then)

AC25. Given I am authenticated and my vehicle has an active pass for the current period, when I open the home screen, then I see ‘X liters remaining of Y’ with the period end time.

AC26. Given a partial dispense just completed, when I refresh (pull-to-refresh), then the displayed remaining quota matches the canonical value from M2 within 2 seconds.

7.3 Module M3 — Fuel Pass (QR Code) Service

M3 turns an Allocation into a Fuel Pass: a signed, time-bound, vehicle-bound QR Code. Payloads are signed with an Ed25519 key held in the platform's key vault; the public key is distributed to all Operator Apps so they can verify offline.

M3-US-01 User Story

As **the System**, *I want* generate one Fuel Pass per Allocation immediately after the weekly allocation job completes, *so that* every eligible citizen has a pass available at the start of the week.

Acceptance Criteria (Given / When / Then)

AC27. Given the `quota.allocated` event is received, when M3 processes it, then it creates a FuelPass row with: vehicle_id, period_start, period_end, allocated_liters, remaining_liters, signed_payload, qr_image_url, state=ACTIVE.

AC28. Given the signing key is rotated, when a new pass is generated, then it is signed with the current `kid` (key id), and the kid is included in the payload header.

AC29. Given M3 fails on a single vehicle, when the batch finishes, then failed vehicles are written to a `dead_letter_passes` table for retry within 15 minutes; aggregate failure rate is reported on the admin dashboard.

M3-US-02 User Story

As **a citizen**, *I want* see my current weekly Fuel Pass on the home screen, *so that* I can simply open the app, hold up the QR, and get fuel.

Acceptance Criteria (Given / When / Then)

AC30. Given I have an ACTIVE pass for the current period, when I open the home screen, then the QR is rendered above the fold within 1 second on a 4G device.

AC31. Given I have no active pass (e.g. vehicle PENDING_VERIFICATION), when I open the home screen, then I see a clear empty state explaining why and what to do next.

AC32. Given the device is offline and a previously fetched pass is still valid, when I open the home screen, then the cached QR is shown with a discreet 'offline' badge and the period end time.

M3-US-03 User Story

As **a citizen without a smartphone**, *I want* receive my Fuel Pass via SMS as a short link, or via USSD as a one-time numeric code, *so that* I can still use the system on a feature phone (Production).

Acceptance Criteria (Given / When / Then)

AC33. Given I dial the NFPS USSD short code and authenticate, when I select ‘Get Fuel Pass’, then the System replies with a 9-digit one-time numeric code valid for 5 minutes, plus the remaining quota.

AC34. Given an SMS request was triggered, when delivered, then it contains a short HTTPS link to a hardened web page that renders the QR; the page expires at the end of the quota period.

AC35. Given the USSD or SMS channel is rate-limited (>3 retrievals in 10 minutes for the same phone), when exceeded, then the System replies with a friendly throttling message in the user’s language.

M3-US-04 User Story

As **a security officer**, *I want* revoke a Fuel Pass on demand (theft, compromise, fraud investigation), *so that* we can stop a redemption that is in flight or imminent.

Acceptance Criteria (Given / When / Then)

AC36. Given I have role `security_officer`, when I revoke pass P with reason R, then the pass state becomes REVOKED and a `pass.revoked` event is published.

AC37. Given an Operator App is online when revocation occurs, when it next validates P, then the central service replies REVOKED and dispensing is blocked.

AC38. Given an Operator App is offline at the moment of validation (Production), when it later syncs, then the dispense is flagged ‘suspect’ and held for review; the citizen is notified.

7.4 Module M4 — Transactions (Station Operator App)

M4 is the runtime of the platform: where the loop closes. The Operator App is the single most performance-critical surface; everything in M4 is designed to be fast, deterministic, and auditable.

M4-US-01 User Story
As **a station operator**, *I want* scan a citizen's Fuel Pass QR code with my Operator App, *so that* I can see in under 2 seconds whether dispensing is authorized and for how many liters.
Acceptance Criteria (Given / When / Then)
AC39. Given the App is logged in to a known station, when I tap 'Scan' and aim at a QR, then the camera focuses and decodes within 800 ms on a mid-range Android. AC40. Given a valid, unrevoked, in-period pass with remaining quota, when the App validates with the central service, then it returns AUTHORIZED with the maximum dispensable liters within 1.5 seconds at the 95th percentile. AC41. Given the central service replies DENIED, when the App receives the response, then it shows a red full-screen result with the reason code (EXPIRED, REVOKED, NO_QUOTA, UNKNOWN) in Amharic and English, and refuses to proceed.

M4-US-02 User Story
As **a station operator**, *I want* enter the actual liters dispensed and confirm the transaction, *so that* the System's record matches the physical dispense.
Acceptance Criteria (Given / When / Then)
AC42. Given an AUTHORIZED scan, when I enter liters L (≤ remaining quota and ≤ pump receipt), then the App calls POST /transactions/confirm with the authorization id and L. AC43. Given the confirm call succeeds, when the App receives 200 OK, then it shows a green confirmation screen with transaction id, liters, and timestamp. AC44. Given I do not confirm within 5 minutes of authorization, when time expires, then the authorization is auto-released by the central service and the citizen's quota is not deducted.

M4-US-03 User Story
As **a station operator (Production)**, *I want* operate the App when the station is offline, *so that* intermittent connectivity does not block fuel sales.
Acceptance Criteria (Given / When / Then)

AC45. Given the App detects loss of connectivity, when I scan a QR, then the App verifies the signature locally using the cached platform public key and reads the embedded period and quota.

AC46. Given the offline check passes, when I confirm a dispense, then the transaction is queued locally, encrypted at rest, with a monotonic local sequence number.

AC47. Given connectivity is restored, when the next sync runs, then queued transactions are uploaded in order; duplicates are rejected by (station_id, local_sequence).

AC48. Given a queued transaction would push remaining quota below zero, when synced, then the central service marks it `partial_overdraw`, accepts the dispense, and flags the case for the operator's station manager.

M4-US-04 User Story

As **an auditor**, *I want* search and trace any single transaction end-to-end, *so that* I can investigate disputes and fraud quickly.

Acceptance Criteria (Given / When / Then)

AC49. Given I have role `auditor`, when I search by transaction id, plate, station, or date range, then I get a paginated list within 2 seconds (P95) for the last 30 days.

AC50. Given I open a transaction, when it loads, then I see: the originating Allocation, the Fuel Pass with its key id, the scan event, the authorization, the dispense confirmation, the operator user id, the device id, and any anomaly flags.

AC51. Given the transaction has a related anomaly case, when I open it, then I am one click away from the case workspace (Production).

7.5 Module M5 — Monitoring, Analytics & Reporting

M5 is read-side only. It never writes to operational tables. It consumes domain events and projects them into purpose-built analytical tables (and, in Production, into OpenSearch and a data warehouse).

M5-US-01 User Story
As **a government administrator**, *I want* see a real-time national dashboard of liters allocated vs liters dispensed, *so that* I can detect emerging shortages or oversupply within hours, not weeks.
Acceptance Criteria (Given / When / Then)
AC52. Given I have role \`gov_viewer\`, when I open the dashboard, then it shows for today and the current week: total allocated, total dispensed, % redeemed, top 10 stations by volume, and a regional heat map. AC53. Given new transactions arrive, when 60 seconds elapse, then the dashboard auto-refreshes and the figures move (Production); MVP refreshes every 5 minutes. AC54. Given I export the view, when I click 'Export CSV', then a CSV download is generated within 10 seconds for the visible filters.

M5-US-02 User Story
As **a fraud analyst (Production)**, *I want* be alerted when a vehicle exhibits anomalous redemption patterns, *so that* I can intervene before quota abuse scales.
Acceptance Criteria (Given / When / Then)
AC55. Given baseline z-scores per category exist, when a vehicle's 7-day rolling redemption count exceeds the threshold, then an Anomaly Case is opened automatically with severity HIGH/MED/LOW. AC56. Given an open case, when I review and act, then I can: dismiss, freeze the pass (M3-US-04), require KYC re-verification, or escalate to law enforcement; every action is audit-logged. AC57. Given a vehicle accumulates 3 dismissed cases in 30 days, when the 4th opens, then it auto-escalates to severity HIGH and notifies the regional admin.

7.6 Module M0 — Core, Admin & RBAC

M0 holds cross-cutting concerns: authentication, authorization (RBAC), audit logging, integration adapters, configuration, feature flags, and platform health.

Default RBAC matrix

Role	Scope	Sample permissions
citizen	Self only	Read own profile, vehicles, passes, transactions.
station_operator	Single station	Scan QR, confirm dispense, view today's log.
station_manager	Single station	Operator perms + edit station hours, view monthly reports.
regional_admin	Region	Read all citizens & transactions in region; no quota policy.
quota_admin	National	Create/edit quota rules, run/cancel allocation jobs.
vehicle_admin	National	Verify vehicles, change categories.
security_officer	National	Revoke passes, freeze accounts, manage anomaly cases.
auditor	National (read-only)	Read all data; no mutations.
gov_viewer	National (read-only)	Read dashboards and exports.
super_admin	Platform	All of the above + RBAC, integrations, feature flags.

8. User Flows

Each flow is described as a numbered, end-to-end sequence. Steps written in italics are System-side; all others are user-side.

8.1 Citizen Onboarding Flow

8. Citizen downloads the NFPS app or visits the web portal.
9. Citizen taps ‘Sign up’, enters Fayda ID and phone number, accepts Terms of Service and Privacy Policy.
10. System validates Fayda format, sends OTP via SMS gateway, displays OTP screen with 5-minute countdown.
11. Citizen enters OTP. If correct, account is created.
12. Citizen is shown the home screen with the prompt ‘Add a vehicle to receive your weekly Fuel Pass’.
13. Citizen taps ‘Add vehicle’, enters plate, region code, engine and chassis numbers.
14. System creates vehicle in PENDING_VERIFICATION (or auto-verifies via Vehicle Registry in Production).
15. Citizen sees ‘Pending verification — usually completes within 24 hours’ with a status timeline.
16. On verification, citizen receives an SMS and an in-app push: ‘Your vehicle is active. Your first Fuel Pass will be ready Monday at 00:00.’

8.2 Weekly Pass Receive Flow

17. (System) Sunday 23:55 EAT — quota engine pre-warms cache for the upcoming period.
18. (System) Monday 00:00 EAT — quota engine runs, creates Allocations for all ACTIVE vehicles.
19. (System) M3 generates a signed Fuel Pass per Allocation; QR images are written to object storage with a public-read short URL.
20. (System) Push notification fan-out begins; SMS fallback for users without push opt-in.
21. Citizen opens the app on Monday morning; the home screen shows the new QR with allocated liters and period end time.

8.3 Fuel Redemption Flow (Citizen + Operator)

22. Citizen drives to a station, opens the app, taps ‘Show Fuel Pass’.
23. Operator opens the Operator App, taps ‘Scan’, the camera activates.
24. Operator scans the QR; the App POSTs to /transactions/authorize with QR payload, station id, device id.
25. (System) M4 verifies signature, expiry, revocation, remaining quota; reserves quota and returns AUTHORIZED with max liters.
26. Operator confirms with the citizen and pumps fuel, then enters the actual liters into the App.
27. Operator taps ‘Confirm Dispense’; the App POSTs to /transactions/confirm with authorization id and liters.

28. (System) M4 commits the dispense, decrements remaining quota, writes the transaction, emits `transaction.completed`.
29. Operator and citizen each see a green confirmation; citizen receives a digital receipt in-app.

8.4 Offline Redemption Flow (Production only)

30. Operator App detects loss of connectivity (no API heartbeat for >10 seconds).
31. Operator App switches into OFFLINE MODE — banner displays at top of screen.
32. Operator scans QR; the App verifies the Ed25519 signature locally using the cached platform public key.
33. If the signature is valid and not in the local revocation list, the App displays AUTHORIZED with the remaining liters from the QR payload's last-known value.
34. Operator dispenses and confirms; transaction is queued locally with a monotonic sequence and a device id.
35. On reconnection, queued transactions are flushed to the central service in order.
36. (System) Late reconciliation: any conflict (e.g. quota already exhausted online) is resolved by 'first commit wins' with the loser flagged for review.

8.5 Admin — Adjusting Quota Policy

37. Admin logs into the back-office, navigates to Quota → Policies.
38. Admin selects the category (e.g. TAXI), clicks 'New version'.
39. Admin sets liters per period and `valid_from` (must be ≥ next period start).
40. System shows a preview: estimated affected vehicles and total liters delta.
41. Admin confirms; the new policy version is saved; the previous version is end-dated.
42. (System) Audit log entry is written; an internal email is dispatched to the policy mailing list.

9. UI / UX Specifications

The product is built for citizens with low literacy, low-end devices, intermittent connectivity, and zero patience for friction. The interface is therefore designed around three principles, in priority order:

9.1 Design Principles

#	Principle	Implication
1	Simple	Each screen has one job. The home screen shows the QR and remaining liters. Nothing else fights for attention.
2	Fast	Time-to-QR ≤ 1 second on app open. Time-to-AUTHORIZED ≤ 1.5 s for the operator (P95).
3	Intuitive	Visual primary, text secondary. The QR is the largest element. Numbers are large; labels are small. Two languages: Amharic, English. One toggle, persisted per user.

9.2 Visual System

Token	Value	Use
Color — Primary (Navy)	#1E3A5F	Headings, primary buttons.
Color — Accent (Green)	#00A651	Success, valid pass, AUTHORIZED.
Color — Warning (Amber)	#F4B400	Pending states, low remaining quota.
Color — Danger (Red)	#C8102E	DENIED, REVOKED, errors.
Background — Canvas	#F2F3F5	App background.
Background — Card	#FFFFFF	Cards and surfaces.
Typography — Heading	Inter / SF Pro / Roboto + Noto Sans Ethiopic	Bilingual support.
Iconography	Lucide	Consistent line icons, 24px grid.
Touch targets	≥ 48 dp	Single-tap reach for low-literacy users.
Contrast	WCAG 2.1 AA	Outdoor sunlight legibility.

9.3 Citizen Mobile App — Screen Inventory (MVP)

#	Screen	Primary action
S01	Splash + language pick	Choose Amharic / English.
S02	Sign-up — Fayda + phone	Submit and request OTP.
S03	OTP entry	Confirm 6-digit code.
S04	Home — Fuel Pass	Show QR, liters remaining, period end.
S05	Add vehicle	Plate, region, engine, chassis.
S06	Vehicle list & detail	View state, archive, edit.
S07	Transactions list	30-day list with station and liters.
S08	Transaction detail	Receipt, station, time, liters, station map.
S09	Profile & settings	Phone, language, sign-out.
S10	Help & FAQ	Static content, contact support.

9.4 Home Screen — Anatomy

The home screen is the most important surface in the entire product. Its anatomy, top-to-bottom:

43. App bar — left: vehicle plate (tap to switch if multiple). Right: bell (notifications).
44. Hero card — large white card containing: ‘Available Fuel Quota’ label, very large number ‘20 Liters’, vehicle icon.
45. QR card — full-width card holding the QR code at minimum 240×240 dp, with green corner brackets.
46. Validity strip — ‘Valid until: Sunday 19:00’ with a thin progress bar showing time remaining in the period.
47. Action row — ‘Find a station’ (opens map), ‘Share via SMS’ (sends short link), ‘Get help’.
48. Bottom navigation — Home / Transactions / Vehicles / Profile.

9.5 Operator App — Screen Inventory (MVP)

#	Screen	Primary action
O01	Login (operator code + PIN)	Authenticate.
O02	Station selector (if multi-pump)	Pick pump.

#	Screen	Primary action
O0 3	Scan	Camera viewfinder, single CTA ‘Scan’.
O0 4	Authorization result	Full-screen GREEN/RED, max liters, vehicle plate.
O0 5	Enter dispensed liters	Numpad, large input.
O0 6	Confirm dispense	Confirm, get receipt id.
O0 7	Day log	List of today’s transactions for this operator.
O0 8	Settings	Sync now, sign out, app version.

9.6 Operator App — Authorization Result Screen

This is the only screen with no decoration: the operator must read it in under 1 second. Anatomy:

- Full-screen background color: GREEN if AUTHORIZED, RED if DENIED. Nothing else competes.
- Single icon at the top (check or cross), 96 dp, white.
- Single very large text below: ‘UP TO 20 L’ on green, or ‘DENIED — EXPIRED’ on red.
- Vehicle plate and last 4 of pass id, smaller, white at 70% opacity.
- One CTA at the bottom: ‘Continue’ on green, ‘Retry / Back’ on red. No other buttons.

9.7 Web Admin Portal — Information Architecture

Section	Pages
Dashboard	Today, This Week, Custom range.
Citizens	List, detail, KYC status, manual override (super_admin only).
Vehicles	List, detail, category change, verification queue.
Stations	List, detail, operators, pumps, hours.
Quota	Categories, rules (with version history), allocations (per period).
Fuel Passes	List, detail (audit trail), revoke (security_officer).

Section	Pages
Transactions	Search, detail, anomaly cases (Production).
Reports	Pre-built reports + custom range CSV export.
Admin	Users, roles, integrations, feature flags, audit log.

10. Backend Architecture (Django)

The same Django codebase ships in MVP and Production. Code lives in a single repository: one Django project, six Django apps (one per module). Architectural change between releases is driven by composition, not code rewrites.

10.1 MVP — Minimum Application Layers

The MVP deliberately collapses layers. Goal: a working pilot in Addis Ababa in ≤ 6 months with 1–2 backend engineers.

MVP Components

Layer	Choice	Why
Web framework	Django 5.x	Mature, batteries-included, native admin, strong ORM.
API layer	Django REST Framework (DRF)	Standard, idiomatic, Token + Session auth.
Auth	JWT (SimpleJWT) + DRF token for service-to-service	Mobile-friendly stateless auth.
Database	PostgreSQL 16, single instance	ACID, JSONB, range types, mature in EAT region.
Cache	Per-process LRU (django-cache `locmem`)	Zero ops; sufficient for pilot load.
Async work	Django management commands + cron (or APScheduler)	Avoids running a broker.
Files / QR images	Local disk under `MEDIA_ROOT`	Operationally simplest.
Search	PostgreSQL `pg_trgm` and `to_tsvector`	No extra service.
Email / SMS	Single adapter (SendGrid / Ethio Telecom SMS)	Adapter pattern enables Production swap.
Logging	Django logging → Sentry	One-line traceability.
Deployment	Single VM, Docker Compose: Django + Postgres + Nginx	Fewest moving parts.
TLS	Caddy or Nginx + Let's Encrypt	Free, automatic.

Layer	Choice	Why
CI/CD	GitHub Actions or Gitea + Woodpecker	Already familiar; automated tests on push.

MVP Layer Diagram (textual)

```
┌───────────────────────────────────────────────────────────┐
│  Mobile App (Android/iOS)  │  Web Portal      │
│  Operator App (Android)    │  Admin Portal    │
└───────────────────────┬───────────────────────────────────┘
                        │  HTTPS (JSON, JWT)
┌───────────────────────▼───────────────────────────────────┐
│             Nginx / Caddy                      │
│    TLS termination, static, reverse proxy      │
└───────────────────────┬───────────────────────────────────┘
┌───────────────────────▼───────────────────────────────────┐
│   Django + DRF (single Gunicorn process)       │
│   apps: core / users_vehicles / quota /        │
│         fuelpass / transactions / analytics    │
└────────────┬───────────────────┬──────────────────────────┘
┌────────────▼───┐  ┌────────────▼──────────────────────┐
│ PostgreSQL │  │  Local disk: MEDIA_ROOT     │
│ single inst│  │  (QR images, receipts)      │
└────────────────┘  └───────────────────────────────────┘
```

10.2 Production — Scaled Architecture

Production is the same Django codebase, deployed differently and with adapters swapped to use real infrastructure. There are no rewrites; configuration and Docker images change.

Production Components

Layer	MVP	Production
App runtime	1 Gunicorn process on a VM	Django on Kubernetes, multi-replica, HPA on CPU + RPS
Edge	Caddy / Nginx	Cloud LB → Ingress → Service mesh (mTLS)
Database	PostgreSQL single instance	Postgres primary + 2 replicas, PgBouncer pool, daily PITR
Cache	locmem	Redis cluster: per-resource TTL, hot keys, surge protection
Async work	Cron jobs	Celery workers + Redis broker, retries, dead-letter queue
Scheduling	OS cron / APScheduler	Celery Beat in HA pair

Layer	MVP	Production
Files / receipts	Local disk	S3-compatible object storage + CDN; signed URLs
Search	Postgres trigram	Postgres + OpenSearch projection for transactions
Event bus	In-process signals	NATS JetStream or Kafka — durable, partitioned by station
Email / SMS / USSD	Single adapter	Multi-provider, weighted failover, delivery receipts
Observability	Sentry + Django logs	Sentry + OpenTelemetry, Prometheus + Grafana, ELK
Secrets	Env vars on the VM	HashiCorp Vault / cloud KMS; Ed25519 signing key in HSM
CI/CD	Build & ship	Build, scan (Trivy, Bandit), test, blue/green, automated rollback
DR	Daily DB backup	Multi-AZ active/passive, RPO ≤ 5 min, RTO ≤ 30 min

Production Layer Diagram (textual)

```
                Citizens / Operators / Admins
                          │
                 ┌────────▼────────┐
                 │   CDN / WAF     │
                 └────────┬────────┘
                 ┌────────▼────────┐
                 │  Cloud LB + TLS │
                 └────────┬────────┘
                 ┌────────▼────────┐
                 │   K8s Ingress   │
                 └────┬───────┬────┘
           ┌──────────▼──┐ ┌──▼───────────┐
           │ Django      │ │ Operator API │
           │ web pods    │ │ (priority)   │
           └──────┬──────┘ └──────┬───────┘
                  │               │
        ┌─────────▼───────────────▼──┐
        │     Redis Cluster          │ (cache + Celery broker)
        └─────────────┬──────────────┘
        ┌─────────────▼──────────────┐
        │  PgBouncer → Postgres      │
        │  (primary + 2 replicas)    │
        └────────────────────────────┘
```

```
Celery workers ──▶ NATS / Kafka ──▶ Analytics projector
                                 └──▶ OpenSearch
                                 └──▶ Object storage (S3) + CDN
```

10.3 Caching Strategy (Production)

Key pattern	TTL	Notes
fuelpass:active:{vehicle_id}	Until period end	Hot path for QR display; invalidated on revoke.
quota:remaining:{pass_id}	60 s sliding	Decremented atomically on dispense; never trusted as source of truth — Postgres is.
pubkey:{kid}	24 h	Platform Ed25519 public key for offline ops apps.
station:lookup:{station_id}	10 min	For Operator App login & metadata.
dashboard:national:today	60 s	Aggregates served via cache aside.
rate_limit:{ip}:{route}	1 min	Token-bucket counters.

10.4 Async Work Catalog (Production)

Task	Trigger	SLA
Weekly allocation job	Beat: Mon 00:00 EAT	Complete in ≤ 30 min for 5M vehicles.
Fuel Pass generation	Event: `quota.allocated`	Complete in ≤ 30 min after allocation.
Push fan-out (FCM/APNs)	Event: `pass.created`	P95 ≤ 5 min.
SMS fallback	Event: `pass.created` & no push	P95 ≤ 10 min.
Analytics projection	Event: `transaction.completed`	End-to-end lag P95 ≤ 60 s.
Anomaly detection sweep	Beat: every 15 min	Per-pass evaluation in ≤ 5 min.
Daily reconciliation	Beat: 02:00 EAT	Finishes by 03:00 EAT.

10.5 File and Receipt Storage

- MVP: Django writes QR PNG to `MEDIA_ROOT/qr/{period}/{pass_id}.png` and serves it through Django’s static handler behind Nginx.

- Production: QR PNG and receipt PDFs are written to an S3-compatible bucket; clients read via short-lived signed URLs (5 min); a CDN fronts the bucket for cacheable receipts.
- All files are tagged with `period`, `vehicle_id`, and `kid`; lifecycle policy deletes objects 90 days after period end.

11. Data Model (logical)

The schema below is the canonical logical model. Concrete Django migrations may add indexes, constraints, and surrogate keys, but field semantics will not change without a versioned ADR.

11.1 Core Entities

Entity	Key fields	Notes
Citizen	id, fayda_id (unique), full_name, phone (unique, +251), language, kyc_state, created_at	1:1 with a Fayda; soft-delete only.
Vehicle	id, citizen_id (FK), plate_number (unique), region_code, engine_no, chassis_no, category_id, state, created_at	States: PENDING_VERIFICATION, ACTIVE, ARCHIVED, BLOCKED.
VehicleCategory	id, code (unique), name_am, name_en, default_liters_per_period	Reference data.
QuotaRule	id, category_id, liters_per_period, period_type, valid_from, valid_until, created_by, version	Versioned; non-overlapping per (category, period).
Allocation	id, vehicle_id, period_start, period_end, allocated_liters, source_rule_id, created_at	Unique on (vehicle_id, period_start).
Adjustment	id, allocation_id, delta_liters, reason, created_by, created_at	Audit-required free text reason.
FuelPass	id, allocation_id (1:1), state, signed_payload, kid, qr_object_url, period_start, period_end, allocated_liters, remaining_liters, created_at, revoked_at, revoke_reason	States: ACTIVE, EXHAUSTED, EXPIRED, REVOKED.
Station	id, code, name, region, lat, lon, is_active	Stations are seeded by admin.
StationOperator	id, station_id, citizen_id (FK), role, pin_hash	PIN required for in-app sign-in.
AuthorizationRequest	id, fuelpass_id, station_id, operator_id, device_id, requested_at, decision, reason_code, max_liters	TTL 5 min.

Entity	Key fields	Notes
Transaction	id, authorization_id, liters, completed_at, station_id, operator_id, status	Statuses: COMPLETED, REJECTED, AUTO_RELEASED, PARTIAL_OVERDRAW.
AuditEvent	id, actor_id, actor_role, action, entity_type, entity_id, payload_json, created_at	Append-only.

11.2 Key Constraints & Indexes

- UNIQUE (vehicle_id, period_start) on Allocation — one allocation per vehicle per period.
- UNIQUE (citizen_id) on Vehicle WHERE state = 'ACTIVE' — a citizen may only have one ACTIVE vehicle of a given category in MVP (relaxed in Production).
- CHECK (remaining_liters >= 0) on FuelPass — enforced at the database level.
- BTREE index on Transaction(completed_at DESC) and (station_id, completed_at DESC).
- GIN index on AuditEvent(payload_json) for fast attribute search.
- Range index on QuotaRule(valid_from, valid_until) using `tsrange`.

12. API Specifications (selected)

All APIs are versioned (`/api/v1/...`), return JSON, authenticate via short-lived JWT (citizen and operator) or session (admin web). Errors follow RFC 7807 (Problem Details for HTTP APIs).

12.1 Citizen — Authentication

```
POST /api/v1/auth/signup/start
Body: { fayda_id: "1234567890123456", phone: "+251911000000", language: "am" }
Returns: 202 { challenge_id, expires_in: 300 }
```

```
POST /api/v1/auth/signup/verify
Body: { challenge_id, otp: "123456" }
Returns: 201 { access, refresh, citizen: { id, name, phone, language } }
```

12.2 Citizen — Vehicles & Pass

```
POST /api/v1/me/vehicles
Body: { plate, region_code, engine_no, chassis_no }
Returns: 201 { id, state: "PENDING_VERIFICATION" }
```

```
GET /api/v1/me/fuelpasses/current
Returns: 200 {
  pass_id, vehicle_plate, allocated_liters, remaining_liters,
  period_start, period_end,
  qr_url,            // signed object URL (5 min TTL in Production)
  signed_payload     // for offline display
}
```

12.3 Operator — Authorization Loop

```
POST /api/v1/operator/transactions/authorize
Headers: Authorization: Bearer <operator_jwt>, X-Device-Id: <uuid>
Body: { signed_payload, station_id, scanned_at }
Returns: 200 {
  decision: "AUTHORIZED" | "DENIED",
  authorization_id, max_liters, expires_at,
  reason_code  // EXPIRED, REVOKED, NO_QUOTA, UNKNOWN, ... when DENIED
}
```

```
POST /api/v1/operator/transactions/confirm
Body: { authorization_id, liters }
Returns: 200 { transaction_id, completed_at, remaining_liters }
```

12.4 Admin — Quota Policy

```
POST /api/v1/admin/quota/rules
Body: { category_id, liters_per_period, period_type: "WEEKLY",
        valid_from: "2025-W41" }
Returns: 201 { rule_id, version, valid_from, valid_until: null }
```

12.5 Error Format


```json
{
  "type": "https://nfps.gov.et/errors/USR-409",
  "title": "Identity already registered",
  "status": 409,
  "code": "USR-409",
  "detail": "This Fayda ID is already linked to an active account.",
  "instance": "/api/v1/auth/signup/start"
}
```

13. Integration Requirements

All external integrations sit behind a thin adapter interface inside the `core` Django app. Adapters expose a stable port; concrete implementations swap between MVP (mock / local) and Production (live).

Integration	MVP	Production	Method
Fayda National Digital ID	Format check + manual override	Live verification API	REST + signed JWT, mTLS
National Vehicle Registry	Manual admin verify	Live lookup	REST or message queue feed
SMS Gateway	Single provider (e.g. Ethio Telecom)	Multi-provider with failover	HTTP API
USSD Gateway	Out-of-scope	Required	HTTP webhook + session manager
Push Notifications	FCM only	FCM + APNs	FCM/APNs SDK
Payment Gateway	Out-of-band	Direct debit / wallet rails	REST + webhook reconciliation
GIS / Maps	Static asset	Live map tiles, station search	Tile server + REST
National Data Warehouse	CSV export only	Daily Parquet drop + streaming	Object storage handoff
Object Storage	Local disk	S3-compatible (MinIO or cloud)	S3 API + signed URLs
Email	SendGrid / SES (transactional)	Same + delivery analytics	SMTP / REST
HSM / Key Vault	OS keystore	HSM-backed Ed25519 signer	PKCS#11

14. Non-Functional Requirements

NFRs are measurable, testable, and tied to release gates. Anything that cannot be measured in CI or in production observability is rewritten until it can.

14.1 Performance

NFR	MVP target	Production target
Citizen home screen QR render	≤ 1.5 s (P95) on 4G	≤ 1.0 s (P95) on 4G
Operator authorize round-trip	≤ 2.0 s (P95)	≤ 1.5 s (P95)
Operator confirm round-trip	≤ 1.5 s (P95)	≤ 1.0 s (P95)
Weekly allocation job	≤ 10 min for 100K vehicles	≤ 30 min for 5M vehicles
Dashboard freshness	≤ 5 min	≤ 60 s
Transaction search (30d window)	≤ 3 s (P95)	≤ 1.5 s (P95)

14.2 Availability & Reliability

NFR	MVP target	Production target
Service availability (read/write APIs)	99.5% monthly	99.95% monthly
Operator authorize endpoint	99.9% monthly	99.99% monthly
RPO (data loss tolerance)	≤ 24 h (daily backup)	≤ 5 min (PITR + replication)
RTO (recovery time)	≤ 8 h	≤ 30 min
Planned downtime	Off-hours announced	Zero-downtime via blue/green

14.3 Scalability

Dimension	MVP	Production
Active citizens	Up to 100,000	Up to 10 million
Active vehicles	Up to 100,000	Up to 5 million
Active stations	Up to 200	Up to 4,000
Peak transactions/sec	Up to 50	Up to 2,000

Dimension	MVP	Production
Daily transactions	Up to 100,000	Up to 10 million

14.4 Security

- All data in transit over TLS 1.3+; HSTS preloaded for the citizen domain.
- All data at rest encrypted (AES-256); database disk-level + sensitive columns at column level.
- Fuel Passes signed with Ed25519; signing key in HSM (Production), filesystem with strict perms (MVP).
- Operator App pinned to platform CA; certificate pinning to defeat rogue Wi-Fi.
- RBAC enforced at the API gateway and re-checked at every service boundary; no implicit admin paths.
- OWASP ASVS Level 2 (MVP), Level 3 (Production).
- Pen test before any release that crosses 25k citizens.
- Vulnerability scanning on every CI build (Trivy for images, Bandit for Python, npm audit for clients).
- Personal data minimization: store only what M1, M2, M3, M4 need; no marketing fields.

14.5 Privacy & Data Protection

- Compliance with the Personal Data Protection Proclamation of Ethiopia (and any superseding law).
- Citizen has the right to view and export their data; deletion is honored as account anonymization (transactions retained for audit).
- PII never appears in logs; trace IDs are used for support cross-referencing.
- Data retention: transactions kept for 7 years; PII purgeable on legal request after audit completion.

14.6 Accessibility & Localization

- WCAG 2.1 AA across all citizen and admin surfaces.
- Bilingual (Amharic, English) at launch; right-to-read order honored where applicable.
- Operator App designed for use under direct sunlight: contrast ratios ≥ 7:1 on result screens.
- All numeric values are localized; ETB currency and Ethiopian Calendar dual-display where appropriate.

14.7 Observability & Operations

- Every API request is traced (W3C Trace Context); P50/P95/P99 latencies, error rates, and saturation are dashboarded per route.
- Structured JSON logging; correlation id required on every cross-service call.
- Health checks: `/healthz` (liveness), `/readyz` (readiness with dependency checks).

- Synthetic monitoring of the citizen sign-up and operator authorize flows every minute (Production).
- Alert on: error rate spikes, P95 SLO burn, queue lag, dead-letter accumulation, signing key access.

15. Security & Compliance Deep Dive

15.1 Threat Model (top 10)

#	Threat	Mitigation
1	QR replay across stations	One-time-use authorization id, 5-minute TTL, server-side decrement before confirm.
2	Forged QR (offline)	Ed25519 signature; operator app verifies with cached public key + revocation delta.
3	Operator collusion (overstating liters)	Pump-side reconciliation in Production; daily station-level audit; anomaly detection.
4	Account takeover	OTP rate limiting, device binding, login alerts via SMS, mandatory re-auth for sensitive actions.
5	Sybil registrations (one citizen, many vehicles)	Hard cap per citizen; admin approval beyond cap; Fayda + chassis cross-checks.
6	Insider data exfiltration	RBAC least privilege; read-only auditors; queries against PII go through a logged proxy.
7	Signing key compromise	HSM, kid rotation, public key distribution channel signed by a CA.
8	DDoS on the operator authorize endpoint	Edge rate limiting, dedicated priority pool, station-id throttling.
9	USSD session hijack (Production)	Short session lifetime, MSISDN binding, no session secrets in URLs.
10	Data poisoning of analytics	Read-side projection isolation, replayable event log, drift checks.

15.2 Controls by Domain

- Identity: OTP, JWT short-lived (15 min) + refresh (7 days), revocation list.
- Cryptography: Ed25519 for QR; AES-256-GCM for column encryption; bcrypt(12) for operator PINs.
- Network: WAF rules; egress allowlist; private subnets for DB and Redis.
- Application: input validation via DRF serializers; output escaping; CSRF on session-auth admin.
- Audit: every state change captured with actor, before/after diff, request id.
- Backups: encrypted, off-region copy at least daily.

16. Acceptance & Test Strategy

16.1 Test Pyramid

Layer	What	Tooling	Coverage gate
Unit	Pure functions, model logic, serializers	pytest + pytest-django	≥ 85% lines in core, M1, M2, M3, M4
Component	DRF view + DB	pytest-django + factory_boy	Every endpoint with happy + 2 sad paths
Contract	Mobile ↔ API, Operator ↔ API	Schemathesis against OpenAPI	Schema break = build fails
End-to-end	Citizen onboard → pass → redeem	Playwright (web) + Maestro (Android)	Run on every release branch
Performance	Authorize loop, allocation job	Locust	P95 budget enforced in CI
Security	SAST, dependency, container	Bandit, Trivy, npm audit	No High / Critical findings

16.2 Definition of Done (per user story)

- Acceptance criteria pass in test.
- Unit + component tests added, suite green.
- OpenAPI schema updated.
- Bilingual UI strings present (Amharic + English).
- Audit log entries written for state changes.
- Observability: latency + error metrics emitted; dashboard updated if a new route.
- Security review for any auth, payment, or PII path.
- Documentation: README section or runbook entry for operations impact.

16.3 Acceptance Pilot Exit Criteria (MVP)

The MVP exits pilot when, for two consecutive weeks, the following are simultaneously true:

- ≥ 95% of pilot citizens receive their weekly Fuel Pass before 06:00 EAT Monday.
- Operator authorize round-trip P95 ≤ 2.0 s during peak hours.
- ≤ 0.5% of transactions are flagged as anomalies that require manual reversal.
- Zero security incidents at severity HIGH or above.
- Sign-off from the government sponsor on the ‘pilot retrospective’ report.

17. Implementation Roadmap

The roadmap follows the three phases described in the concept deck. This SRS is the binding scope reference for Phase 1 and the architectural basis for Phases 2 and 3.

17.1 Phase 1 — Pilot (Months 0–6)

- Pilot in selected areas (Addis Ababa + 1 region).
- Vehicle scope: private cars and taxis.
- Core platform, QR system, station Operator App, basic admin portal.
- Public awareness and station operator training.
- Exit gate: pilot exit criteria above.

17.2 Phase 2 — National Rollout (Months 6–18)

- Expand to all regions.
- All vehicle categories (incl. public transport, government, commercial).
- Strengthen infrastructure & integrations (Fayda live, Vehicle Registry live, Payments).
- Performance monitoring & continuous optimization.
- USSD/SMS channel and Service Center workflows live.

17.3 Phase 3 — Optimization (Months 18+)

- AI-driven demand forecasting and policy planning.
- Advanced analytics, anomaly case workflow, public open-data API.
- Policy refinements (region/zone/woreda overrides).
- Continuous improvement and innovation cycles.

17.4 Resource Snapshot (indicative)

Phase	Backend	Mobile	QA	DevOps	Other
Phase 1 (Pilot)	2	2 (1 Android, 1 iOS)	1	1 (part-time)	1 PM, 1 Designer
Phase 2 (Rollout)	4	3 + 1 web	2	2	1 PM, 1 Designer, 1 Sec. Eng.
Phase 3 (Optimize)	4 + 1 Data	Same	2	2	+1 ML Eng., +1 Data Analyst

18. Risks & Mitigations

#	Risk	Likelihood	Impact	Mitigation
R1	Fayda integration not yet ready when MVP ships	M	H	Adapter pattern with mock; format-check + manual override; switch on adapter swap.
R2	Station connectivity gaps	H	H	Production offline mode with signed-QR verification; replay reconciliation.
R3	Operator usability collapse under load	M	H	Single-screen authorization result; sub-second decode; operator pilot training.
R4	Forged or shared QR codes	M	H	Signature + one-time-use + station/device fingerprinting + anomaly detection.
R5	Data privacy backlash	M	H	Privacy-by-design; minimum data; clear citizen-facing privacy notice; DPIA before pilot.
R6	Quota policy political pressure (rapid changes)	H	M	Versioned policy with valid_from; transparent admin audit trail.
R7	Scaling bottlenecks at Monday 00:00 EAT (allocation peak)	M	M	Pre-warmed cache; batched DB writes; burst-capable workers.
R8	Insider data abuse	L	H	RBAC, read-only auditors, query proxy with logging, regular access reviews.
R9	Vendor lock-in via cloud-native services	M	M	Use S3-compatible APIs, Postgres + Redis (open) — runs on TeleCloud or any cloud.
R10	Pilot regional politics (selection of 1st region)	M	M	Sponsor-led selection; clear exit criteria; symmetrical onboarding playbook.

19. Appendices

19.1 Appendix A — Sample Fuel Pass Payload (signed)

```
{
  "v": 1,                         // payload version
  "kid": "nfps-2025-Q4",        // signing key id
  "pid": "fp_01J9X3Z4...",      // pass id
  "vid": "veh_01J9X3Z4...",     // vehicle id
  "plate": "AA-12345",
  "liters": 20,
  "period": ["2026-04-27T00:00:00+03:00", "2026-05-03T23:59:59+03:00"],
  "issued_at": "2026-04-27T00:00:42+03:00"
}
// Wire format: base64url(payload) + "." + base64url(ed25519_signature)
```

19.2 Appendix B — Error Code Catalog (excerpt)

Code	HTTP	Meaning
USR-409	409	Identity already registered.
USR-403	403	Account locked or KYC failed.
VEH-409	409	Vehicle already registered to another citizen.
QUO-409	409	Overlapping quota policy version.
QUO-422	422	Adjustment would result in negative quota.
PASS-410	410	Fuel Pass expired.
PASS-403	403	Fuel Pass revoked.
TXN-409	409	Authorization already consumed or expired.
AUTH-401	401	Missing or invalid token.
AUTH-403	403	Insufficient permissions for this action.

19.3 Appendix C — Reference Mapping to Concept Deck

Concept slide	SRS section
Slide 1 — Cover & promise	§2 Executive Summary
Slide 2 — The Challenge	§2 Vision; §14.4 Security; §15 Threat model
Slide 3 — Our Solution (Register / Allocate / Generate / Redeem / Monitor)	§6 Architecture; §7 Modules

Concept slide	SRS section
Slide 4 — Alignment with national priorities	§2.2 Strategic Alignment
Slide 5 — How it works for citizens	§8.1 Onboarding; §8.3 Redemption
Slide 6 — System architecture	§6 Architecture; §10 Backend
Slide 7 — Key features	§7 Modules; §14 NFRs
Slide 8 — Benefits	§2.3 Expected impact
Slide 9 — Roadmap	§17 Implementation Roadmap
Slide 10 — Expected impact / Commitment	§2.3, §16.3 Pilot exit